

AUSTIN LIEN

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Education

University of California, Santa Cruz

Sep. 2023 – June 2027

Bachelor of Science in Computer Engineering, Minor in Computer Science

Santa Cruz, CA

Relevant Coursework

- Computer Systems Design
- Machine Learning & Data Analysis
- Data Structures & Algorithms (C, C++)
- Parallel and Concurrent Programming
- Computer Architecture
- Computer Networks

Work Experience

Undergraduate Researcher – LLM Bias & Narrative Evaluation Lab

April 2025 - Present

Prof. Markus Eger, UCSC

Santa Cruz, CA

- Built a llm-bias audit pipeline using ModernBERT and DistilBERT to quantify sentiment deltas and embedding similarity across text variants at scale.
- Optimized inference performance with batched GPU/CPU-aware transformer models, enabling efficient high-volume analysis across large text corpora.
- Engineered a robust data ingestion layer that processes unstructured sources (txt, csv) into structured reports featuring sentiment arcs, similarity metrics, and metadata.
- Developed an automated analysis suite that generates summary tables and statistical visualizations (histograms, boxplots) to surface systemic bias patterns.

ACM AI Research – Vision-Language Reliability

September 2025 - Present

Student Researcher, UCSC

Santa Cruz, CA

- Built Python based multimodal AI evaluation pipelines to benchmark GPT-4V, LLaVA, and Bard across 190+ hallucination cases, generating quantitative model quality and safety metrics.
- Implemented prompt engineering and LLM inference experiments (CoT, self-correction, adversarial prompts), analyzing failure modes in reasoning, OCR, and image/text interference for large vision-language models.
- Developed automated data pipelines with version-controlled workflows (Git/GitHub), producing reproducible analyses and proposing GenAI improvements such as counterfactual data augmentation and RAG-style conditioning.

Projects

Sentinel Spend | *Python, SQL, Scikit-learn, Streamlit, RapidFuzz, pdfplumber*

- Developed an End-to-End ETL Pipeline in Python to ingest and standardize unstructured data from PDF statements into a unified SQLite schema.
- Automated Maverick Spend Detection via SQL to identify non-compliant vendor pricing by reconciling transaction data against Master Contract tables.
- Integrated ML Anomaly Detection using an Isolation Forest model to flag high-risk transactions based on amount, frequency, and merchant risk.
- Designed a Streamlit Dashboard to visualize forensic audit results and geographic spend patterns, translating data into actionable business insights.

Slouching Slugs | *Python, OpenCV, Mediapipe, Gemini API, NumPY*

- Developed real-time posture detection app analyzing webcam feeds with 95% slouch detection accuracy across 500+ test frames.
- Achieved intelligent real-time alerts (200ms latency), integrated Google Gemini with OpenCV + MediaPipe landmark detection.
- Utilized NumPy for matrix operations & image processing to detect significant pixel deviations across multiple image frames.

GrocerView | *React Native, Typescript, Python, ChromaDB*

- Built a cross-platform React Native (Expo) barcode-scanning app with manual entry, ingredient parsing, alerts, and ownership insights using Open Food Facts & USDA; persisted user preferences with Zustand + AsyncStorage.
- Developed a FastAPI backend that standardizes ingredients via parallel queries to PubChem, FoodOn, and Open Food Facts, merging synonyms to improve search recall.
- Designed a multi-source research pipeline (PubMed, OpenAlex, PMC) with deduplication, relevance ranking, and licensed full-text ingestion into ChromaDB.
- Implemented RAG summaries with quote verification and traceable citations, integrating regulatory sources (FDA GRAS, JECFA, Codex, NIH ODS) and caching to reduce LLM calls.

Technical Skills

Awards: ICPC Pacific Northwest Div 2. (1st - California, 5th - Regional)

Languages: Python, Java, Javascript, C, Risc-V Assembly, C++, SQL

Technologies/Frameworks: AWS, GCP, Linux, Git, OpenCV, MediaPipe, Pandas, Matplotlib, PyTorch,